



Folding DNA

The 2006 paper by Paul WK Rothemund that started it all, and created the art of DNA Origami. <https://dgit.in/aug20-17>



Nanobots fighting cancer

This animated video shows how DNA nanobots use thrombin to cut off blood supply to tumours. <https://dgit.in/aug20-18>

DNA origami

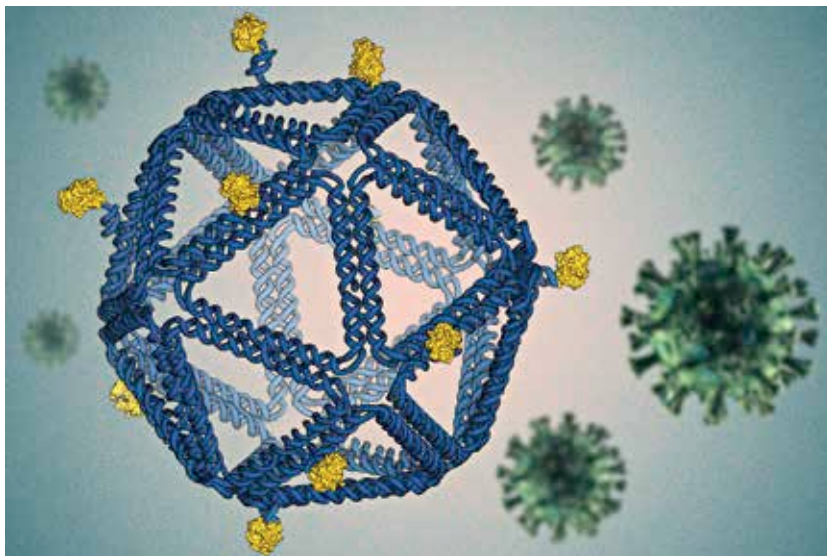
Engineering DNA into machines

Aditya Madanapalle | aditya@dgit.in

In 2006, Paul Rothemund, a professor of Bioengineering, Computing and Mathematical

Sciences, and Computation and Neural Systems from Caltech published a paper in Nature called “Folding DNA to create nanoscale shapes and patterns”. The idea was to use DNA strands to create nano structures of any arbitrary shape. These included disks, stars and squares and even words, all as small as 6nm. Combining multiple 6nm “pixels” together could be used to form any kind of even more complex shapes. The process of fabrication for these nanostructures is helped by the natural tendency of the base pairs in DNA to find suitable partners. To recap high school biology, the adenine always pairs with the thymine (A-T) and the cytosine always pairs with the guanine (C-G). Say you want to make a particular shape, all you need to do is compute the sequence of bases or purines in a strand, after which the DNA assembles or folds itself into the desired shape. This process is called DNA origami. Just like a single sheet of paper is used to make origami, DNA origami mostly uses a single strand of DNA.

Since that time, there have been many advances in the field and many potential applications have opened up. Research from Harvard allowed for the creation of complex 3D shapes, while MIT researchers developed increasingly sophisticated algorithms.



An HIV-like particle created using DNA origami that provokes an immune response

The research in the field is mainly around finding new methods of synthesis, or fabrication, finding newer and better algorithms for creating the shapes, and exploring potential medical applications. All kinds of things can be attached to the base DNA shapes, from nanoparticles, to proteins, fluorescent molecules and carbon nanotubes. These methods allow researchers to make all kinds of small machines and some even with electrical circuits. These large and complex molecules can be used to make biosensors, deliver drugs, store memory and perform artificial photosynthesis. For the last application, it is necessary to recreate 20 protein subunits in particular positions and orientations, which DNA origami is able to do. DNA origami is used in a number of fields now and is a cutting edge tool in medicine, chemistry, biology and material science. Since 2005, DNA origami has become faster and DNA synthesis has become cheaper, so the technique is more accessible to researchers now.

DAEDALUS

In 2011, the researchers at MIT made advances in automating the process of creating the 3D shapes by using algorithms to accurately predict the complex ways in which DNA strands, bend, twist and fold during the formation of the shapes. The approach also allows researchers to estimate the amount of stress and strain at any point in the shape, which allows them to evaluate their usefulness for a particular purpose. The idea at that time was to give the software a 3D shape based on which the algorithm would automatically provide a DNA sequence. This vision took five years to realise.

In 2016, MIT researchers developed an algorithm called DNA Origami Sequence Design Algorithm for User-defined Structures (DAEDALUS) named after the greek craftsman who designed the labyrinth. This algorithm could create any given shape out of DNA including complex ones with one or more holes in them, such as a torus. The process was entirely auto-

Credit: MIT